

TESD 1800 – Software Development (4 Credits)

Course Description

In this course, students will explore the Software Development Life Cycle. Students will test code, practice searching, sorting, building data structures, using generic objects and collections, and asynchronous processing. Students will be exposed to modern project management styles.

Course Objectives

- Practice project management techniques.
- Design feature specification for software.
- Apply Code Design patterns.
- Design code using common data structures.
- Explore the Software Development Lifecycle.

Course Outline

- Introduction to Objects
- Inheritance and Polymorphism
- Exception Handling
- Abstract Classes and Interfaces
- Introduction to Graphics
- Event Driven Programming
- Binary Input/Output
- Final Project

Textbook & Reading Materials

Introduction to Java Programming and Data Structures, 13th Ed (Lifetime access) (ISBN: 9780138123352)

Assignments & Assessments

Orientation
Course Introduction and Standards
Meet Your Instruction Team
Rules of the Lab
CS Code of Conduct Policy
Submitting Assignments in Canvas
Taking a Screenshot
Programming Standards
GitHub Standards
Writing Standards
Exam Standards
Syllabus Agreement
Course References
Student Data Files
Check Exercises
Book Videos
Module 1 Introduction and Competencies
Exercise 9-1: Planning
Exercise 9-1: Implementation
Exercise 9-7: Planning
Exercise 9-7: Implementation
Chapter 9 GitHub
Chapter 10 - Reading Day
Exercise 10-3: Planning
Exercise 10-3: Implementation
Exercise 10-7: Planning
Exercise 10-7: Implementation
Chapter 10 GitHub
Module 1 Exam
Checkpoint Meeting Module 1
Student Acknowledgement Statement Module 1
Module 2 Introduction and Competencies
Chapter 11 - Reading Day
Exercise 11-1: Planning
Exercise 11-1: Implementation
Exercise 11-3: Planning
Exercise 11-3: Implementation
Chapter 11 GitHub
Module 2 Exam
Checkpoint Meeting Module 2
Student Acknowledgement Statement Module 2
Module 3 Introduction and Competencies
Chapter 12 - Reading Day
Exercise 12-3: Planning
Exercise 12-3: Implementation
Exercise 12-15: Planning
Exercise 12-15: Implementation

Chapter 12 GitHub
Module 3 Exam
Checkpoint Meeting Module 3
Student Acknowledgement Statement Module 3
Module 4 Introduction and Competencies
Chapter 13 - Reading Day
Exercise 13-1: Planning
Exercise 13-1: Implementation
Exercise 13-7: Planning
Exercise 13-7: Implementation
Exercise 13-11: Planning
Exercise 13-11: Implementation
Chapter 13 GitHub
Module 4 Exam
Checkpoint Meeting Module 4
Student Acknowledgement Statement Module 4
Module 5 Introduction and Competencies
Installing JavaFX
First Time Using JavaFX
Chapter 14 - Reading Day
Exercise 14-15: Planning
Exercise 14-15: Implementation
Exercise 14-28: Planning
Exercise 14-28: Implementation
Chapter 14 GitHub
Chapter 15 - Reading Day
Exercise 15 - Listeners: Planning
Exercise 15 - Listeners: Implementation
Exercise 15 - Animation: Planning
Exercise 15 - Animation: Implementation
Chapter 15 GitHub
Module 5 Exam
Checkpoint Meeting Module 5
Student Acknowledgement Statement Module 5
Module 6 Introduction and Competencies
Chapter 16 - Reading Day
Exercise 16-1: Planning
Exercise 16-1: Implementation
Exercise 16-21: Planning
Exercise 16-21: Implementation
Chapter 16 GitHub
Module 6 Exam
Checkpoint Meeting Module 6
Student Acknowledgement Statement Module 6
Module 7 Introduction and Competencies
Chapter 17 - Reading Day
Programming 17-1

Programming 17-3

Programming 17-7

Programming 17-14 & 17-15

Chapter 17 GitHub

Module 7 Exam

Checkpoint Meeting Module 7

Student Acknowledgement Statement Module 7

Module 8 Introduction and Competencies

Final Project: Proposal

Final Project: Pitch

Final Project: Code

Final Project: GitHub

End of Course Survey

Course Completion Checkpoint Meeting

End of Course Student Acknowledgement Statement

Subject to change. Please consult your Canvas course for the most current instructions and updates.

Classroom Hours

Mo, Tu, W, Th

8:00 AM - 11:00 AM

12:00 PM - 3:00 PM

4:00 PM - 7:00 PM

Friday

8:00 AM - 11:00 AM

12:00 PM - 3:00 PM

For a full list of course hours visit: [Course Schedule](#)

Instructor Contact Information

Jason Adams — jadams@stech.edu

Dallin Savage — dsavage@stech.edu

Nathan Merrill — nmerrill@stech.edu

Office Hours: By appointment

Email is the preferred method of communication; you will receive a response within 24 hours during regular business hours.

Canvas Information

Canvas is where course content, grades, and communication will reside for this course.

- Canvas URL: stech.instructure.com
 - For Canvas passwords or any other computer-related technical support contact Student Services.
 - For Regular Hours and Weekdays call (435) 586 - 2899
 - For After Hours and Weekends call (435) 865 - 3929 (Leave a message if no response)
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Policies

Course Grading

All assignments in this course require 100% score with unlimited submissions. All quizzes require a minimum score of 80%. High School Power School Grades: Quarter student grades will be determined by student progress percentage. Faculty will use the higher percentage of either 1) quarter progress, or 2) cumulative progress for the current training plan year.

Grade Scale: The following grading scale will be used to determine a letter grade.

A : 94 - 100%	A- : 90 - 93%	B+ : 87 - 89%	B : 83 - 86%
B- : 80 - 82%	C+ : 77 - 79%	C : 73 - 76%	C- : 70 - 72%
D+ : 67 - 69%	D : 63 - 66%	D- : 60 - 62%	F : 0 - 59%

Course Policies

You are required to keep your progress and attendance at 67% minimum. You must complete this program within 150% estimated program length. You are permitted one 15 minute break every 90 minutes. If you take more than one break in a 90 minute period or your break lasts longer than 15 minutes, your attendance will be penalized. 10 consecutive absences will lead to being withdrawn from the program. Please notify your instructors about absences as soon as possible. If absence is due to illness, please email your instructors prior to end of day.

Cell Phone/Electronics: Cell phones cannot be used during class time. You may bring your personal computers to class. You must be on topic in the lab while clocked-in.

Industry Environment: Computer science typically is very sedentary. This means you may sit at a desk for long hours. Be sure to move and get what exercise you can.

Iron County High School Students: Your progress grade will be submitted every Friday to PowerSchool. You must maintain a progress percentage of 67% or over per quarter to stay in the course.

Additional Information

InformaCast Statement: Southwest Tech uses InformaCast to ensure the safety and well-being of our students. In times of emergency, such as weather closures and delays, this app allows us to promptly deliver notifications directly to your mobile devices. To stay informed and receive real-time updates, we encourage all students to sign up for notifications. Your safety is our priority, and staying connected ensures a swift response to any unforeseen circumstances. More information and directions for signing up are available at: <https://stech.edu/emergency-notifications/>

Internet Acceptable Use Policy: The student is expected to review and follow the Southwest Technical College Internet Safety Policy at: <https://stech.edu/students/policies/>

Student Code of Conduct Policy: The student is expected to review and follow the Southwest Technical College Student Code of Conduct Policy at: <https://stech.edu/students/policies/>

Accommodations: Students with medical, psychological, learning, or other disabilities desiring accommodations or services under ADA, must contact the Student Services Office. Student Services determines eligibility for and authorizes the provision of these accommodations and services. Students must voluntarily disclose that they have a disability, request an accommodation, and provide documentation of their disability. Students with disabilities may apply for accommodations, based on an eligible disability, through the Student Services office located at 757 W. 800 S., Cedar City, UT 84720, and by phone at (435) 586-2899. No diagnostic services are currently available through Southwest Technical College.

Safety and Building Maintenance: The College has developed and follows a variety of plans to ensure the safe and effective operation of its facilities and programs. The following plans are available online: 1) Facilities Operations and Maintenance Plan; 2) Technical Infrastructure Plan; and 3) Health and Safety Plan.

Withdrawals and Refunds: Please refer to the Southwest Technical College Refund Policy at: <https://stech.edu/students/policies/>

Any high school or adult student, who declares a technical training objective is eligible for admission at Southwest Technical College (Southwest Tech). Program-specific admissions requirements may exist and will be listed on the Southwest Tech website. A high school diploma or equivalent is not required for admission but is mandatory for students seeking Title IV Federal Financial Aid.

Non-Discriminatory Policy: Southwest Technical College affirms its commitment to promote the goals of fairness and equity in all aspects of the educational enterprise, and bases its policies on the idea of global human dignity.

Southwest Tech is committed to a policy of nondiscrimination. No otherwise qualified person may be excluded from participation in or be subjected to discrimination in any course, program or activity because of race, age, color, religion, sex, pregnancy, national origin or disability. Southwest Technical College does not discriminate on the basis of sex in the education programs or activities that it operates, as required by Title IX and 34 CFR part 106. The requirement not to discriminate in education programs or activities extends to admission and employment. Inquiries about Title IX and its regulations to STECH may be referred to the Title IX Coordinator, to the Department of Education, and/or to the Office for Civil rights.

If you believe you have experienced discrimination or harassment on our campus, please contact the Title IX Coordinator, Cory Estes: cestes@stech.edu, (435) 865-3938.

For special accommodations, please contact the ADA Coordinator, Cyndie Tracy: ctracy@stech.edu, (435) 865-3944.

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